

Next, we will create the sentence matrix associated with each question, as follows:

```
def question2sequence(ques):
    tokens = get_tokens(ques)
    rows = [] #represents the sentence embedding matrix
    #Greedy search for tokens
    for token in tokens:
        assert (token in w2v_wordmap), word + "not
        found in w2v_map"
        rows.append(w2v_wordmap[token])

    if len(tokens) < MAX_QN_LEN:
        #question is too short
        #we need to pad the question up to max sequence
        length
        j = MAX_QN_LEN - len(tokens)
        word = _UNK
        while j > 0:
            rows.append(w2v_wordmap[word])
            j = j - 1
    return rows
```

Now, let's convert this sentence matrix into a fixed-length vector representation of the question. Given a sequence of

input words (this constitutes the question sentence), we now pass this sequence through a recurrent neural network (RNN) and create an output sequence. We can use either vanilla RNNs, gated recurrent units (GRU) or long short term memory (LSTM) units for creating a fixed-length representation from a given input sequence. Given that LSTMs have been quite successfully used in many of the NLP tasks, we decided to use them to create the fixed-length representation of the question.

```
import tensorflow as tf
question1 = tf.placeholder(tf.float32, [N, 1, q1, D],
    'question1')
lstm_cell = tf.nn.rnn_cell.BasicLSTMCell(num_lstm_units)
value, state = tf.nn.dynamic_rnn(lstm_cell, question1,
    dtype=tf.float32)
```

While RNN generates an output for each input in the sequence, we are only interested in the final aggregated representation of the input sequence. Hence, we take the output of the LSTM at the last time step and use it as our sentence representation. Note that the last time step corresponds to the last word in the sentence being fed to the LSTM. Hence, the LSTM output corresponds to an aggregated representation of the current word and all

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